ANTHONY MOLICA

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EDUCATION

Adrian College, Bachelor of Arts in Data Analytics, Adrian, MI — August 2024 - Expected May 2026

Minor in Computer Science

- Member of the NCHA All-Academic Team

St. Lawrence University, Bachelor of Science in Data Science, Canton, NY — August 2022 - May 2024

- Member of the Academically Elite All-ECAC All-Academic Team - Winter 2023

TECHNICAL SKILLS

Languages: SQL, Python (Pandas, NumPy, Scikit-learn, Matplotlib), R, Java, Excel

Data Analysis: Exploratory Data Analysis, Feature Engineering, Regression, Classification

Machine Learning: Logistic Regression, Decision Trees, Random Forest, XGBoost

Statistics: Hypothesis Testing, ANOVA, Chi-Square, ROC-AUC, Model Evaluation

Data Handling: Web Scraping, Data Cleaning, Data Visualization

PROJECTS

NHL Shot Data Analytics Capstone (Python) — August 2025 – December 2025

- Developed an Expected Goals (XG) Model using XGBoost and Random Forest 2023-2025 short-level data
- Engineered features, identified key predictors, conducted exploratory data analysis, and performed model testing and evaluation
- Identified distance, angle, and individual skill to be the strongest predictors of goal probability, demonstrated that nonlinear models significantly outperform linear models, where 7% of shot attempts become goals

NHL Playoff Performance Prediction – Detroit Red Wings — Spring 2024

- Analyzed 2023-2024 NHL play-by-play data using R to evaluate team discipline, defensive impact, power-play efficiency, and scoring trends through data wrangling, visualization, and comparative league analysis
- Built shot plots and team performance metrics to identify key areas for improvement and translated findings into actionable recommendations for Detroit Red Wings playoff qualification
- Quantified the statistical gaps between Detroit and playoff teams, prioritizing special team efficiency and shot suppression to improve postseason probability

HelpHub Web Platform (Web & Data Integration Project) — Fall 2024

- Designed and deployed a volunteer charity matching platform using HTML, CSS, JavaScript, and GitHub Pages
- Integrated the OpenWeatherMap API and refined usability through wireframing and iterative design improvement
- Delivered a fully functional live web application with real-time data integration, improving user engagement and demonstrating full-stack development and API implementation skills

ATHLETIC EXPERIENCE

Adrian College Division 3 NCAA Men's Ice Hockey, *Defense* — August 2024 – Present

St. Lawrence University Division 1 NCAA Men's Ice Hockey, *Defense* — August 2022 – March 2024

Maryland Black Bears, Lincoln Stars, and New Jersey Titans Junior A Hockey, *Defense* — August 2019 - June 2022

- Robertson Cup Champion for the 2021/2022 season with New Jersey Titans
- All teams: Applied video/data analysis using Instat to evaluate team and individual performance
- All teams: Volunteered monthly to support youth hockey programs

WORK EXPERIENCE

Distinct Finish LLC, *Contracted Labor,* Byron Center, MI — May 2020 – August 2024

- Provided interior/exterior painting services, surface preparation, priming, and precision finishing
- Enhanced customer service skills and developed teamwork in a fast-paced environment